CITY EDUCATION INNOVATION WEEK (CEIW) 2026

# subAI

*from the Bisaya word 'subay' — to track*

## Track Every Clogged Drain from Report to Recycled Bottle

AI-Driven Drainage Triage & Verified Circular Plastic Recovery for Cagayan de Oro City

**“Subay the drain. Save the bay.”**

**CHALLENGE AREA A: WASTE TO WORTH | STAGE OF DEVELOPMENT: WORKING PROTOTYPE**

**TEAM PEW PEW • 1st Year BS Computer Science, University of Science and Technology of Southern Philippines (USTP – CDO)**

Team Leader: Danlord Farell Soriano | Lead Developer: Justine Roy P. Salvador | UI/UX: John Kurt Montero

Research: Shawn Michael Hitalada | Pitch: Mark Anthony Bacus | Faculty Coach: Isaiah Marc Postrero

# Problem: 80 Tonnes of Daily Plastics Choking Urban Drains

**450 t**
Municipal solid waste generated per day across CDO (2024 CLENRO/WaCT baseline)

**80 t**
Post-consumer plastics every day (18% of waste stream)

**11%**
Of plastics escape collection as unmanaged leakage

**12**
Urban natural waterways & outfalls receiving leaked trash

## How Plastic Leakage Directly Triggers Urban Flash Flooding:

- Severe Hydraulic Choking: Discarded single-use sachets and PET beverage bottles accumulate at surface curb inlets, catch basins, and trash grates along the 11-km Bitan-ag Creek, crippling hydraulic conveyance by 60%–75%.
- 30–45 Minute Inundation: During convective cloudbursts exceeding 35 mm/hour (PAGASA Yellow/Orange alerts), runoff rapidly backs up into knee- to waist-deep (0.5m–1.2m) floods across Carmen, Lapasan, Kauswagan, Cogon, and Bulua.
- Linear Disposal Dead-End: Municipal response ('Hapsay Sapa') is slow and reactive—deploying only AFTER streets flood. Dredged plastics are hauled unsegregated and unweighed to the Pagalungan Landfill, accelerating landfill saturation.

*Sources: CLENRO 2024 WACS / Waste Flow Diagram; CDRRMD 2024–2025 Flood Inundation Logs; DENR-EMB Region 10 Landfill Compliance Audits.*

# Evidence: Official CLENRO 2024/2025 Waste Inventory

RPD CLENRO <rpdclenro@gmail.com>
to me

4:18 PM (4 hours ago)

DANLORD FARELL A. SORIANO
Team Leader, USIP Cagayan de Oro

Dear Sir,

Good day!

This is in response to your request for waste management data in support of your entry in the Green Innovation Pitching Competition of the City Education Innovation Week 2026.

Please be informed that the attached data for Items 1 to 3 are based on the available records for 2024, which are being used as reference data for 2025, as the available information for both years remains consistent.

| ARANGAY / JURISDICTI | AREA (Has) | POPULATION | 10% TRANS. | TOTAL POP. | ANNUAL (0.58kg) | BIO (54%) | REC (21%) | RES (23%) | SPE (2%) |
|---|---|---|---|---|---|---|---|---|---|
| OF CAGAYAN DE ORO (C | 57,851 | 741,617 | 74,162 | 815,779 | 157,000 | 84,780.17 | 32,970.07 | 36,110.07 | 3,140.01 |
| rmen (Primary Flood Epice | 956.65 | 78,612 | 7,861 | 86,473 | 16,642 | 8,986.77 | 3,494.85 | 3,827.70 | 332.84 |
| swagan (Highway Bottlen | 512.53 | 41,356 | 4,136 | 45,492 | 8,755 | 4,727.74 | 1,838.56 | 2,013.66 | 175.10 |
| pasan (Bitan-ag Creek Out | 227.26 | 40,498 | 4,050 | 44,548 | 8,573 | 4,629.65 | 1,800.42 | 1,971.89 | 171.47 |
| amaman-an (Upstream Ba | 731.92 | 37,775 | 3,778 | 41,553 | 7,997 | 4,318.36 | 1,679.36 | 1,839.30 | 159.94 |
| lua (Western Commercial | 661.21 | 36,293 | 3,629 | 39,922 | 7,683 | 4,148.94 | 1,613.48 | 1,767.14 | 153.66 |

## 157,000 MT / Year

Total CDO solid waste generated (0.58 kg/day per capita across 815,779 population).

## 32,970 MT / Year (21%)

Recyclable Fraction (90.3 MT/day): Valuable polymers that subAI intercepts at the drain.

## 3,495 MT / Year (Carmen)

Carmen alone generates 16,642 MT/yr (3,495 MT recyclables), making it the #1 flood epicenter.

**Direct Proof: Recyclables are 21% of CDO waste. Catching them at catch basins keeps millions of pesos in the circular loop.**

# Who It Serves: Target Users & Citywide Beneficiaries

## 1. Citizens & Commuters

**~815,779 Population**

Act as distributed crowd-reporters submitting geo-tagged obstruction photos with client-side canvas compression (<100ms) on low-bandwidth networks.

## 2. Municipal Maintenance

**~120 Active Staff**

CLENRO 'Hapsay Sapa', CDRRMD Oro Rescue, barangay tanods, and campus maintenance receive prioritized work orders, upload after-photos, and log scale weights (kg).

## 3. Recyclers & MRFs

**Accredited Partners**

Local junk shop cooperatives and barangay Materials Recovery Facilities accept sorted, scale-verified plastic batches and issue signed digital receipts.

## Direct Beneficiaries Across Cagayan de Oro City:

- High-Risk Flood Barangays: Residents and business owners in Carmen, Lapasan, Kauswagan, Cogon, and Bulua gain flood-resilient streets, safe transit along Sayre Highway, and protected storefronts.
- Academic & Medical Centers: Students, patients, and staff across USTP, Capitol University, Liceo de Cagayan, Xavier University, and Northern Mindanao Medical Center (NMMC) avoid hazardous street isolation.
- Coastal Ecosystem of Macajalar Bay: Intercepts thousands of kilograms of single-use polymers at storm drains before runoff discharges into marine waters, protecting coastal fishing livelihoods.

# Solution: Track Every Drain, Report to Recycler

## 1. Report
**/report**

Photo capture, <100ms client compression, Gemini Vision AI scan, OSM GPS zone pinning.

## 2. Triage
**/review**

Live triage queue, duplicate detection, and objective 0–5 priority scoring.

## 3. Execute
**/tasks**

Digital work order, after-cleanup photo evidence, scale weights (kg) by material stream.

## 4. Handover
**/dashboard**

Transfer sorted batches to MRFs / recyclers with signed digital receipts.

## 5. Reconcile
**/dashboard**

Authoritative real-time mass balance calculation: 100% auditable, no greenwashing.

**Strict Reconciled Mass Balance Formula (Zero Greenwashing):**

**Total Collected (kg) = Accepted for Recycling + Residual to Landfill + Awaiting Handover**

Unweighed estimates are NEVER counted toward recycled mass. Every gram requires an authoritative scale receipt.

# Circular Economy Model: Rethink, Recover, Recycle, Repurpose, Reduce

## 1. Rethink

**Drains as Interceptors**

Re-envisions clogged storm drains not merely as municipal flood liabilities, but as decentralized passive collection points that capture polymers before entering Macajalar Bay.

## 2. Recover

**Point-of-Extraction**

Displaced polymers and silt are segregated immediately at the drain inlet into discrete fractions (PET bottles, caps, flexible polyolefins, inert silt) rather than contaminated.

## 3. Recycle

**PET to Pellets**

Weighed, clean PET beverage bottle batches are routed directly to certified mechanical recyclers for industrial flake and pellet production.

## 4. Repurpose

**Films to Eco-Bricks**

Low-density multilayer flexible packaging and food sachets are diverted to local partner upcycling enterprises for durable eco-bricks and paving blocks.

## 5. Reduce

**Hotspot Source Policy**

Granular spatial hotspot maps reveal exactly where packaging litter accumulates, giving City Hall and barangays empirical evidence to ban target single-use plastics.

### How subAI Designs Out Waste & Keeps Materials in Use (Competition Mandate):

- Keeps Materials in Use: Diverts plastics from 450-year landfill decomposition in Pagalungan into perpetual circular recycling and construction applications.
- Regenerates Natural Systems: Restores the natural hydraulic runoff capacity of urban creeks and protects Macajalar Bay marine habitats from microplastic degradation.
- Resource Cascading (Rain Bank): Built-in runoff divert models retention capacity for roof rainwater harvesting, supplying water for campus irrigation and street cleaning.

# Technology & AI Component with Strict Ethical Guardrails

## 1. Core Technology Stack

Framework: Next.js 16 (React 19, TypeScript 5)
Database & Auth: Firebase Cloud Firestore + Security Rules
Mapping & GIS: Leaflet + OpenStreetMap + Nominatim reverse geocoding
Architecture: Mobile-first PWA, zero app-store friction
Design Standards: Flat solid color tokens, strictly zero gradients

## 2. Gemini AI Vision Engine

Model: Google Gemini 2.5 Flash / 3.5 Flash-Lite
Function: Ultra-fast (<1.5s) multimodal scan of obstruction photos
Detection: Identifies PET bottles, flexible films, silt, organics, rubble
Constraints: Compressed base64 (≤960×960 px, <200 KB)
Output Schema: Structured JSON: severity score + category chips

## 3. Safety & Ethical Guardrails

Human-in-the-Loop: AI outputs are advisory suggestions; humans override
Data Privacy (RA 10173): No facial/biometric scanning; coordinates sanitized
Fail-Safe Design: Seamless manual fallback if Gemini API quotas drop
Anti-Hallucination: AI never guesses mass (kg) or declares drain safety
Physical Verification: Digital scales prove actual kilograms, not AI estimates

# Feasibility: PHP 85,000 Year-1 Pilot & Revenue Model

## Year-1 Citywide Pilot Budget: PHP 85,000

• Cloud Infrastructure (PHP 15,000): Vercel serverless hosting, Firebase Cloud Firestore, custom domain, and Gemini API quota buffers.
• Hardware & Verification Tools (PHP 25,000): 3 heavy-duty industrial digital hanging/platform scales, waterproof ruggedized clipboards, and field PPE for maintenance personnel.
• Community & Barangay Onboarding (PHP 30,000): CLENRO & CDRRMD workflow integration workshops, barangay tanod training, and durable QR code hotspot plaques at drainage grates.
• Contingency & Equipment Maintenance (PHP 15,000): Scale calibration, physical equipment maintenance, and emergency response reserves.

## Sustainability & Revenue Streams

• 1. B2G & Campus SaaS Licensing: White-label maintenance deployments for city engineering offices, neighboring LGUs (El Salvador, Opol, Tagoloan), and university campuses.
• 2. Recyclables Arbitrage: Aggregating certified, scale-verified post-consumer PET plastic batches sold in bulk directly to industrial recyclers at premium market rates.
• 3. EPR Compliance Audits (RA 11898): Supplying auditable plastic waste diversion credits to FMCG corporations mandated to recover their packaging footprints.
• Strategic Partners: City Local Environment and Natural Resources Office (CLENRO), CDRRMD Cagayan de Oro, City Education and Development Office (CEDO), Association of Barangay Captains (ABC), USTP, and CDO junk shop cooperatives.

# Expected Impact: 12-Month Quantifiable Targets

## 5,000 kg

### Waste Diverted from Drains

At least 5,000 kg of solid waste intercepted from curb inlets, including ≥2,000 kg PET bottles and ≥1,200 kg flexible films certified via scale receipts into circular recycling.

## 48h → <6h

### Drain Clearance Time Cut

60% reduction in average drainage obstruction resolution time during storms, eradicating chronic street ponding across 12 identified city flood hotspots.

## 2,500+ / 800+

### Active Users & Work Orders

2,500+ active citizen/barangay crowd-reporters, 800+ completed digital work orders, and 100% reconciled mass-balance audit trail.

**How Results are Verified (No Estimates, Authoritative Truth):**

1. Scale receipts (digital weight slips in kg) signed by accredited recyclers • 2. Server-verified GPS coordinates and before/after photos • 3. Automated JSON mass-balance audit exports for city officials and competition judges.

# Alignment with UN Sustainable Development Goals (SDGs)

### SDG 11: Sustainable Cities

Target 11.5 & 11.6: Eliminates street flood ponding, protects vulnerable residential communities, and modernizes municipal solid waste and drainage systems.

### SDG 12: Responsible Consumption

Target 12.5: Substantially reduces solid waste disposal through rapid source interception, rigorous material sorting, and verified circular MRF handovers.

### SDG 14: Life Below Water

Target 14.1: Prevents land-based plastic debris and microplastics from entering Bitan-ag Creek and emptying into Macajalar Bay.

### SDG 13: Climate Action

Target 13.1: Strengthens municipal adaptive resilience against severe monsoon precipitation by pairing drainage unblocking with rainwater retention modeling.

### SDG 9: Industry & Innovation

Target 9.4: Deploys modern, low-bandwidth, AI-assisted civic infrastructure that integrates mobile PWA technology with municipal sanitation workflows.

### SDG 17: Partnerships for the Goals

Target 17.17: Builds collaborative cross-sector partnerships between City Hall (CLENRO/CDRRMD), higher education (USTP), barangays, and private recycling cooperatives.

**Challenge Area A: Waste to Worth — Solid waste segregation & diversion; plastic, organic/food & e-waste reduction; recycling & tracking.**

# Team Pew Pew — 1st Year BS Computer Science, USTP CDO

**Danlord Farell Soriano**
**Team Leader**
Full-Stack Architecture & Systems Coordination

**Justine Roy P. Salvador**
**Lead Developer**
Next.js, Firebase Cloud Firestore & Leaflet GIS Integration

**John Kurt Montero**
**UI/UX Designer**
Mobile PWA Experience & Responsive Design Tokens

**Shawn Michael Hitalada**
**Concept Paper + Research**
Data Synthesis, CLENRO/CDRRMD Analytics

**Mark Anthony Bacus**
**Presentation + Pitch**
Pitch Delivery, Q&A Strategy & Video Walkthrough

**Faculty Coach: Isaiah Marc Postrero**
Department of Computer Science, University of Science and Technology of Southern Philippines (USTP) – Cagayan de Oro Campus

# The Ask: Support to Pilot and Scale Across Cagayan de Oro

Empowering Kagay-anon to transform municipal drainage from a flood hazard into a circular resource.

### 1. Pilot Access & City Integration

Official partnership with CLENRO & CDRRMD to integrate subAI with 'Hapsay Sapa' teams across 12 high-risk flood bottlenecks in Carmen, Lapasan, Kauswagan, Cogon, and Bulua.

### 2. Barangay & Civic Onboarding

Support from the Association of Barangay Captains (ABC) to conduct orientation workshops for barangay tanods and install QR reporting markers at curb inlets.

### 3. Recycler Network Formalization

Coordination with accredited junk shop cooperatives and barangay MRFs to formalize weigh-in receipts and digital transfer agreements.

### 4. Seed Incubation & Mentoring

PHP 85,000 pilot capital for digital scales, field equipment, and cloud hosting, alongside technical incubation to prepare for municipal adoption.

## “Subay the drain. Save the bay.”

Team Pew Pew • USTP Cagayan de Oro • Contact: Danlord Farell Soriano